

- ① 오답.md 어제 오답부터 재풀이
- ② 신규는 공부계획 ID대로
- ③ 채점: solve-exam-prep 로 풀이 대조

떠올릴 개념

- center $\omega_c = 1/\sqrt{LC}$
- 대역폭 β , $Q = \omega_c / \beta$

문항 목차 (총 4문항)

- [1] 14.18 — 필터·공진
- [2] 14.21 — 필터·공진
- [3] 14.35 — 필터·공진
- [4] 14.38 — 필터·공진

14.18 For the bandpass filter shown in Fig. P14.18, find
PSPICE (a) ω_o , (b) f_o , (c) Q , (d) ω_{c1} , (e) f_{c1} , (f) ω_{c2} , (g) f_{c2} , and
MULTISIM (h) β .

596 Introduction to Frequency Selective Circuits

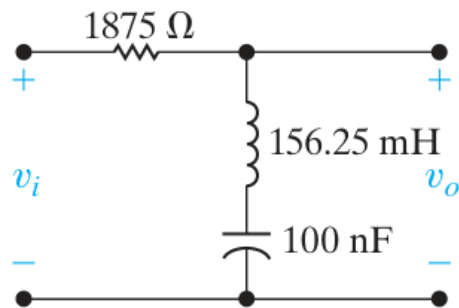
Figure P14.18

14.21 Design a series RLC bandpass filter (see Fig. 14.19[a]) with a quality of 8 and a center frequency of 50 krad/s, using a $0.01\ \mu\text{F}$ capacitor.

- a) Draw your circuit, labeling the component values and output voltage.
- b) For the filter in part (a), calculate the bandwidth and the values of the two cutoff frequencies.

- 14.35** For the bandreject filter in Fig. P14.35, calculate
(a) ω_o ; (b) f_o ; (c) Q ; (d) β in hertz; (e) ω_{c1} ; (f) f_{c1} ;
(g) ω_{c2} ; and (h) f_{c2} .

Figure P14.35



14.38 For the bandreject filter in Fig. P14.38, calculate
PSPICE (a) ω_o ; (b) f_o ; (c) Q ; (d) ω_{c1} ; (e) f_{c1} ; (f) ω_{c2} ; (g) f_{c2} ; and
MULTISIM (h) β in kilohertz.

Figure P14.38

